

Vamshi Krishna Racharla

Data Engineer

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PROFESSIONAL SUMMARY

Data Engineer with 6+ years of experience designing, building, and owning enterprise data platforms across financial services and telecom organizations. Deep hands-on background in Azure (Data Factory, Databricks, Synapse Analytics, ADLS Gen2), dbt, Python, and SQL, with a strong record of designing scalable data pipelines, ingestion frameworks, and dimensional data models that power analytics and business decision-making. Skilled at improving data quality, monitoring, and pipeline reliability, and at partnering directly with analytics, engineering, and business stakeholders to take projects from design through deployment and long-term support.

EXPERIENCE

GCP Data Engineer, *American Express*

Feb 2026 – Present

- Designed BigQuery partitioning and clustering strategies, cutting query costs by 30% while improving performance across large-scale analytical datasets.
- Lead delivery of large-scale ingestion pipelines and cloud data platform capabilities, partnering with marketing, MDM, and analytics teams on production reporting solutions.
- Orchestrate Airflow workflows for scheduling and monitoring, and drive CI/CD automation (GitHub, Jenkins) across dev, QA, and production environments.

Data Engineer, *Verizon*

June 2023 – Feb 2026

- Architected end-to-end Azure pipelines — ADLS Gen2, Azure Data Factory, Azure Databricks, Azure Synapse Analytics for network data processing and enterprise reporting.
- Implemented CDC-based incremental dbt pipelines processing only changed records, cutting data processing volume by 90% and improving pipeline runtime by 60% versus full-load processing.
- Implemented dbt tests and Python/PySpark validation checks to catch schema drift, missing records, and reconciliation gaps before data reached downstream consumers.
- Modeled star-schema datasets for network and customer analytics, documenting definitions, ownership, and lineage alongside the analytics team.
- Tuned Snowflake and BigQuery query performance through partitioning, clustering, and join optimization.

Big Data Engineer, *Credit Suisse Bank*

May 2022 – Mar 2023

- Developed SCD Type 1 and Type 2 dimensional models in Snowflake to manage current and historical customer attributes, providing 100% historical traceability for financial reporting.
- Delivered batch and real-time pipelines with Kafka, AWS Glue, Lambda, and Redshift, and optimized warehouse usage for financial reporting workloads.
- Migrated legacy Oracle, SQL Server, and DB2 workloads to cloud platforms, automating infrastructure with Terraform and applying governance and encryption controls for regulated data.

Data Engineer, *Wipro*

June 2019 – Dec 2021

- Built batch and near-real-time pipelines with Kafka, Spark Streaming, Python, and Airflow, structured into star and snowflake reporting models.
- Automated ETL processes with Python and Airflow, improving operational efficiency by 25%, and added validation and reconciliation checks that caught missing, duplicate, and inconsistent records before downstream use.
- Maintained lineage and documentation for regulated reporting datasets, supporting audit and compliance needs.

PROJECT

- Pulse360 Data Platform** [Github](#) — Built a Python-based synthetic customer data platform with automated validation for duplicate records, missing IDs, invalid event types, and transaction amounts, achieving a 98.03% data-quality score and quarantining invalid records.
- Built automated data-quality tests with 8 Pytest cases covering data generation and validation scenarios, achieving a 100% test pass rate.

SKILLS SUMMARY

- Azure & Cloud Data Platform:** Azure Data Factory, Azure Databricks, Azure Synapse Analytics, ADLS Gen2, Azure SQL, GCP (BigQuery, Dataflow, Cloud Composer), AWS (S3, Redshift, Glue, Lambda)
- Languages & Transformation:** Python, SQL, T-SQL, dbt, PySpark, Spark SQL
- Data Modeling & Architecture:** Star/snowflake schema, fact & dimension tables, SCD Type 1/2, surrogate keys, CDC processing, data quality & validation, data lineage
- Orchestration & DevOps:** Apache Airflow, Kafka, Hadoop, Hive, Git, GitHub, Jenkins, Terraform, Docker, Kubernetes, CI/CD
- Databases & Analytics:** Snowflake, BigQuery, Redshift, SQL Server, PostgreSQL, Oracle, Tableau, Power BI

EDUCATION

Southern Arkansas University — MS, Computer and Information Science

May 2023

Sree Chaitanya College of Engineering — B.Tech, Computer Science

June 2019